RETRONIX
The Solutions for Connected World

RENESAS

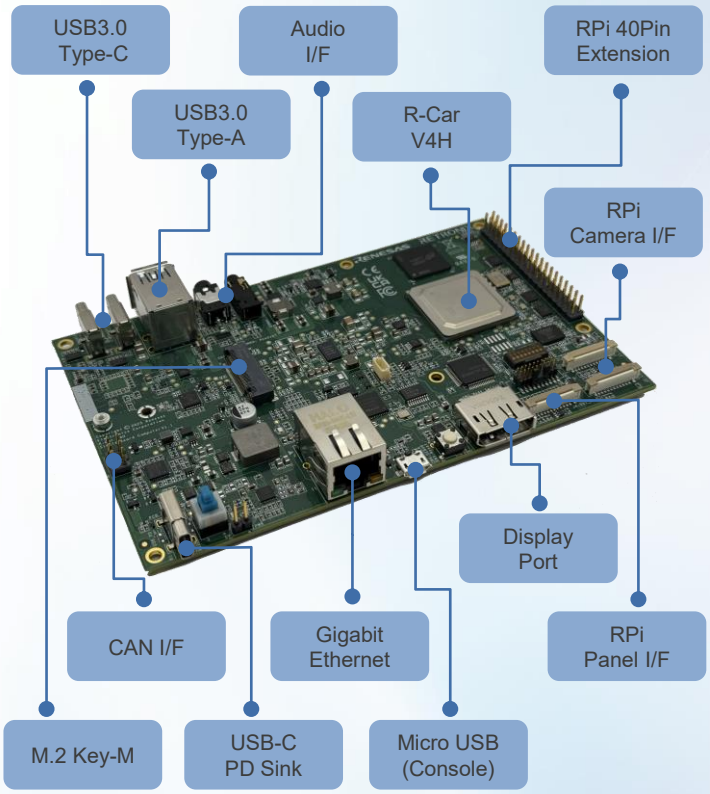
Integrated development environment for AI services and low-cost compact board

R-Car V4H Sparrow Hawk

Core benefits

- **Ready to run OSS based software and start development**
  - Get started instantly with OSS based SW supporting Yocto & Debian, available FoC (Free Of Charge) with just Click though agreement
- **Unleash 30TOPS AI power in a compact board design**
  - Experience up to 30 TOPS range AI performance on a compact board (146 × 90 mm) and fully compatible with Raspberry Pi® HAT.
- **Affordable and accessible hardware for everyone**
  - Low-cost board available through global distributors and High-volume production ensuring stable supply and large inventory

HW Features



SW Features

Supported

Not supported yet

Debian 13	Yocto v5.0	Zephyr (Dom0)	Linux (DomD/U)	Android (DomA)

Linux 6.18

u-boot 2026.04

Xen Hypervisor

Renesas Electronics
www.renesas.com

Board variants	Basic Kit		Complete Kit	
DDR	8G	16G	8G	16G
Sparrow Hawk	✓	✓	✓	✓
Argon FAN			✓	✓
Micro USB cable			✓	✓
32GB Micro SD card			✓	✓

HW Features	
CPU	4x Arm® Cortex®-A76 (81K DMIPS) 3x Arm® Cortex®-R52 (25K DMIPS)
CNN-IP	30 TOPS DENSE
GPU	IMG A-Series AXM-8-256 (150 GFLOPS)
DRAM	LPDDR5 6400Mbps 8GB / 5500Mbps 16GB
Flash Memory	64MB QSPI
Camera I/F	2x MIPI CSI Camera
Display	1x DP, 1x DSI
Ethernet AVB	1 Port (1Gbps)
Debug Serial	2 Port
Audio	In/Out, In
PCIe	1x M.2 Key-M (x2 lane)
USB3.0	2x USB type-A, 2x USB type-C
CAN-FD	2 Port
Removal Media	1x MicroSD
Extensions	Raspberry Pi® 40-Pin GPIO header
Power	USB PD 20V

SW Features	
Yocto BSP	Yocto Project 5.0.16
Debian based BSP	Debian 13
Linux kernel	V6.18.20
u-boot	2026.04
Trusted Firmware-A	V2.14.0

Please see the details here：
https://rcar-community.github.io/Sparrow-Hawk/

- Board / SW info
- Getting started
- Get the board

Please see the details here：
https://www.retronix.com.tw/en/product_sbc.html

- Web page of Retronix Technology Inc.
- Board info
- Board UM

Where to buy：
http://www.shimafuji.co.jp/en/products/2207

- Web page of Shimafuji Electric Incorporated
- Contact : info@shimafuji.co.jp

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V1.02